Bethune Energy

Supply Well Feasibility Study Overview

Executive Summary

Bethune Energy is a proposed approximately 500 MW natural gas-fueled electricity generating facility near the Town of Bethune in Kershaw County, South Carolina, being developed by Beaufort Rosemary. Beaufort Rosemary commissioned Alliance Consulting Engineers, Inc., with groundwater specialists HydroSource Associates, P.C., to evaluate whether on-site wells could supply the Project's anticipated water needs and how pumping could affect nearby wells.

The study concludes that the available information generally supports a groundwater supply of approximately 150,000 gallons per day, with negligible impact to existing documented wells within three miles of the site based on a 24-hour pumping scenario. Final well design, production capacity, number and location of wells, and effects on nearby wells will be confirmed through test drilling, pumping tests, and water-quality sampling, and will be coordinated with and reviewed by the South Carolina Department of Environmental Services (SCDES).

Analysis Methodology

The study reviewed published federal and state groundwater studies, geologic maps, and available well records from the SCDES Coastal Plain Well Inventory. It identified documented public-supply, industrial, agricultural, and residential wells (if publicly available) within three miles of the site, reviewed potential sources of groundwater contamination, and estimated water-level changes in the aquifer from a single simulated Project well. The environmental records search did not identify any nearby facility considered a concern for groundwater quality at the site; water quality will be confirmed by sampling.

The study scenario assumed a single well near the center of the site pumping approximately 104 gallons per minute continuously for 24 hours, the rate needed to supply 150,000 gallons in a day. This represents a conservative approach relative to intermittent pumping at the same rate. Actual pumping schedules will be developed through final engineering, field testing, and coordination with SCDES.

Because underground conditions vary, the calculations were run using two published values for how readily water moves through the aquifer (transmissivity of 3,000 and 17,000 gallons per day per foot), taken from past pumping tests of two existing wells in the area. These are regional estimates, not site-specific test results. With the lower value, water-level changes are larger near the pumping well but extend a shorter distance; with the higher value, they are smaller near the well but extend farther.

Potential Water Supply

The study identifies the McQueen Branch Aquifer, a deeper sand aquifer, as the preferred water source. Existing wells in the area believed to use this aquifer, including public-supply and industrial wells, have reported yields of 100 to 500 gallons per minute. The Project's need of about 104 gallons per minute is at the low end of that range, which supports the feasibility of the proposed supply on a preliminary basis.

Based on available records, the study identified seven of the documented wells, five of which are domestic wells that have publicly available information, as drawing from the shallower Crouch Branch

Aquifer. As a simplifying assumption, the calculations treated all nearby wells as drawing from the same deeper aquifer as the Project well. Published studies report a clay layer separating the McQueen Branch and the Crouch Branch Aquifers in this area. Through test drilling, the Project will evaluate the thickness and continuity of the clay layer to determine the effects drawdown in each aquifer has on the other.

Potential Effects on Nearby Wells

Pumping a well can temporarily lower the groundwater level around it while it operates, an effect called drawdown; water levels recover when pumping slows or stops. A change in water level does not, by itself, mean that a neighboring well would stop producing water. Whether a change matters for a particular well depends on the well's depth, where its pump is set, and how much water stands above the pump.

The study estimated water-level changes after 24 hours of continuous pumping at the 25 documented wells within three miles of the site. The well inventory is based on state records, and some private household wells may not appear in those records. For 22 of those documented wells, the estimated change is 0.04 foot (about one-half inch) or less, and for 14 of them it is less than 0.01 foot. The largest estimated change at a household well is about 0.12 foot (about 1½ inches). The largest estimated change overall, about 0.44 foot (about five inches), is at the nearest documented well approximately 0.86 mile from the assumed Project well location. These calculations support the report's overall findings, and the study characterizes these effects as negligible.

Study Basis and Design

The findings in the study support the feasibility of supplying the Project with approximately 150,000 gallons per day of groundwater based on the conservative study approach used. Final well locations, the number of wells, pumping schedules, and facility water demand may be refined as engineering and field-testing progress. It is expected that SCDES will consider effects on the aquifer, on other water users including public water supplies, and on streams and other surface waters in their reviews. Any changes in design affecting groundwater demand or pumping conditions will be re-evaluated through the applicable technical and SCDES review processes.

The Project is being designed to recycle process water where it is practical to minimize groundwater withdrawals. The 150,000 gallons per day figure is a planning value that includes margin; typical daily use is expected to be lower. Additionally, on-site water storage is being sized to help reduce groundwater withdrawals further.

The Project expects to install more than one supply well so that plant operations do not depend on any single well. The wells would share the plant's water demand rather than add to it: combined withdrawals from all wells would be limited to approximately 150,000 gallons per day. The effects of the final well layout will be evaluated as part of the SCDES review process.